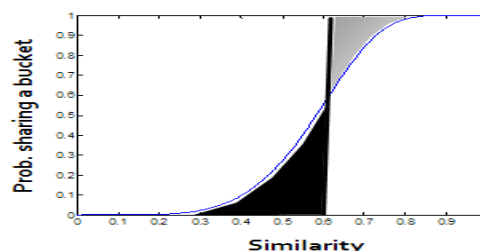


Final Examination
Introduction to Massive Data Analysis

May 25, 2016

1. (12%) Explain the following terms briefly.
 - (a) Zipf 's Law
 - (b) Curse of dimensionality
 - (c) TF.IDF (Term Frequency times Inverse Document Frequency)
2. (10%) Yes/no questions. You don't have to justify the answers.
 - (a) Locality-sensitive hashing (LSH) is a near-neighbor search method.
 - (b) Often, your data looks like a collection of sets, and the objective is to find pairs of sets that have a relatively large fraction of their elements in common. This is the frequent itemsets problem.
 - (c) Calculate the expected number of occurrences of the events you are looking for, on the assumption that data is random. If this number is significantly larger than the number of real instances you hope to find, then you must expect almost anything you find to be TRUE.
 - (d) PageRank was used to simulate where Web surfers, starting at a home page, would tend to congregate if they followed randomly chosen outlinks, and this process were allowed to iterate many times.
 - (e) The algorithm HITS tries to measure two different aspects of importance. Authorities are those pages that contain valuable information. Hubs are pages link to places where the information can be found. The solution is essentially an iterated matrix–vector multiplication, just like PageRank's. However, the existence of dead ends or spider traps does not affect the solution to the HITS equations in the way they do for PageRank.
3. (6%) Show that e (natural logarithm) is the limit of $(1+1/x)^x$ as x goes to infinity
4. (8%) In LSH with r rows and b bands, let columns C_i and C_j having similarity t .
 - (a) The probability that some row in band unequal is _____, the probability that that at least one band identical is _____.
 - (b) Picking the parameters r and b to get the best S-curve as shown in the figure below, where the grey area is the _____ rate and the black area is the _____ rate.



5. (8%) Prove that $\Pr[h_\pi(C_1) = h_\pi(C_2)] = \text{sim}(C_1, C_2)$, the probability that the minhash function for a random permutation π of rows produces the same value for two sets equals the Jaccard similarity of those sets.
6. (8%) Consider a Web site gathering statistics on how many unique users it has seen in each given month. The universal set is the set of logins for that site, and a stream element is generated each time someone logs in. The Flajolet-Martin algorithm can solve this problem if we do not have enough space to maintain the set of elements seen so far. Briefly describe the Flajolet-Martin algorithm and show how it works.
7. (6%) Below are the three questions of the PageRank solution. Briefly describe the answers and reasons.
 - (a) Does this formula converge?
 - (b) Does it converge to what we want?
 - (c) Are results reasonable?
8. (6%) In the block-stripe parallel implementation of PageRank, show that the computational cost per iteration of Power method is $|M|(1+\epsilon) + (k+1)|r|$, where k is the number of block.
9. (8%) BFR is a variant of k -means designed to handle very large (disk-resident) data sets. And BFR is an efficient way to summarize clusters, where the memory requirement is of $O(\text{clusters})$ instead of $O(\text{data})$. Discuss the main concept with three sets of points which we keep track of, and describe the reason why memory space is $O(\text{clusters})$.
10. (6%) The adwords problem which is listed below, fill in the blanks.
 - (a) Given: 1). A set of _____ by advertisers for search queries. 2). A _____rate for each advertiser-query pair. 3). A _____ for each advertiser. 4). A _____ on the number of ads to be displayed with each search query.
 - (b) Google did not simply order ads by the amount of the bid, but by the amount they expected to receive for display of each ad. That is, the value of an ad was taken to be the product of the _____ and the_____.
11. (12%) In the general case, the worst competitive ratio of the BALANCE (many bidders) on-line algorithm is $1-1/e \approx 0.63$.
 - (a) (3%) Gives the worst case example with N advertisers of the same budget B , and NB queries in N rounds.
 - (b) (2%) What's the optimum revenue in the off-line solution?
 - (c) (7%) Show that the revenue of BALANCE is $BN(1-1/e)$, and thus the

competitive ratio is $1-1/e$.

12. (10%) One way of predicting the blank values in a utility (rank) matrix is to find two long, thin matrices Q and P , $R \approx Q \cdot P^T$. Since the matrix product gives values for all user-item pairs, that value can be used to predict the value of a blank in the utility matrix.
 - (a) (2%) The intuitive reason this method makes sense is that often there are a relatively small number of issues (that number is the “thin” dimension of Q and P) that determine whether or not a user likes an item. Lists some possible issues mentioned in the classroom.
 - (b) (2%) In the Netflix Challenge, the reconstruction error (sum of squared errors) should be as smaller as possible, what’s the theoretical solution that the minimum can be guaranteed? How this solution is corresponding to Q and P in (a)?
 - (c) (2%) To solve overfitting problem which may appear in the optimization process we introduce the concept of regularization, give the minimization model with Q , P and regularization rules.
 - (d) (4%) In the implementation for the Challenge, the Stochastic Gradient Descent (SGD) is indispensable, give the reason why and show how it works.

Bonus problems, choose any two.

13. (8%) Here is an example of a tiny Bloom filter. It uses an array of 10 bits and two independent hash functions f and g . We want to test membership in a set S of three elements, so we hash each of the three elements using both f and g , and we set to 1 any bit that any of the three elements is hashed to by either of the hash functions.

When a new element x arrives, we compute $f(x)$ and $g(x)$, and we say x is in the set S if both $f(x)$ and $g(x)$ are 1. Assume x is not in the set S . What is the probability of a false positive; i.e., the probability of saying that x is in S . Identify this probability, correct to three decimal places, in the list below.

(a).780 (b) .220 (c) .121 (d) .178.

14. (8%) Suppose we are using the DGIM algorithm to estimate the number of 1's in suffixes of a sliding window of length 40. The current timestamp is 100, and we have the following buckets stored:

End Time	100	98	95	92	87	80	65
Size	1	1	2	2	4	8	9

Suppose the query asks us to estimate the number of 1's in the 6 most recent elements. What are the largest and smallest number of 1's that could be the correct answer, given the information above? What would the DGIM algorithm estimate? Identify the true statement in the list below.

- (a) The minimum possible number of 1's is 5.
- (b) The maximum possible number of 1's is 5.
- (c) The estimated number of 1's is 1.
- (d) The maximum possible number of 1's is 3.

15. (8%) Here is a collection of twelve baskets. Each contains three of the six items 1 through 6.

$\{1, 2, 3\}$ $\{2, 3, 4\}$ $\{3, 4, 5\}$ $\{4, 5, 6\}$
 $\{1, 3, 5\}$ $\{2, 4, 6\}$ $\{1, 3, 4\}$ $\{2, 4, 5\}$
 $\{3, 5, 6\}$ $\{1, 2, 4\}$ $\{2, 3, 5\}$ $\{3, 4, 6\}$

Suppose the support threshold is 4. On the first pass of the PCY Algorithm we use a hash table with 11 buckets, and the set $\{i, j\}$ is hashed to bucket $i \times j \bmod 11$.

- (a) By any method, compute the support for each item and each pair of items.
- (b) Which pairs hash to which buckets?
- (c) Which buckets are frequent?
- (d) Which pairs are counted on the second pass of the PCY Algorithm?